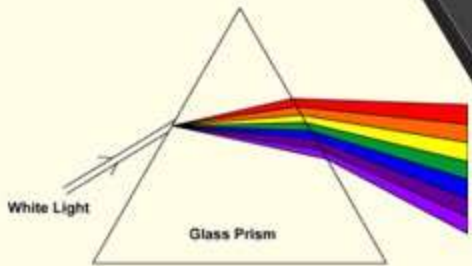


PHYSICS

MORE ABOUT MOTION...

MODULE - 3





KINDS OF MOTION



CAR ON A HIGHWAY

Object covers equal distance
in equal intervals of time.

UNIFORM MOTION

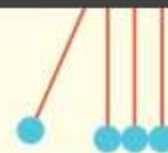


CARS ON A BUSY STREET

Object covers unequal distance
in equal intervals of time.

NON-UNIFORM MOTION





UNIFORM MOTION - EXAMPLES



WATERFALL



CAR ON A LONELY HIGHWAY



BALL ROLLING



SOLDIERS MARCHING IN PARADE

Object covers equal distance in equal intervals of time.



NON-UNIFORM MOTION - EXAMPLES



AEROLANE WHILE TAKING OFF



PEOPLE ON DAILY JOGGING



A CAR ON A CROWDED STREET



A BOWLER THROWING A BALL

Object covers unequal distance in equal intervals of time.



WHAT DO WE UNDERSTAND IN ALL THESE EXAMPLES ?



All These examples show that , different objects may take different time to cover a distance. Some objects move fast while some move slowly.

SLOW AND FAST?



Distance covered = 5 km

Time = 15 min



Distance covered = 5km

Time = 5 min

Some objects move fast than others. This means they cover more distance than other in a particular time



SLOW AND FAST?



**RATE OF MOTION = 8
KILOMETERS IN ONE SECOND**



**RATE OF MOTION = 8
CENTIMETERS IN 1 SECOND**

So, what is the rate of motion called as ?



RATE OF MOTION SPEED



$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

SI

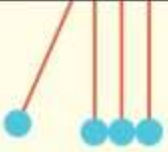
meter per second
 ms^{-1}

SPEED - DEFINITION

The rate at which an object an object moves. It is also defined as the distance travelled per unit time.



DASHBOARD OF A CAR EXPLAINS SPEED AND DISTANCE



ODOMETER



- Measures the distance covered by the vehicle
- Measuring unit is km

SPEEDOMETER



- Measures the velocity of the vehicle in general
- Measuring unit is km/hr

TACHOMETER



- Measure the rotation of the shaft of the engine motor.
- It is measured in rpm (rotations per minute)





QUESTION

Which of the following instrument is used in a vehicle for measuring the total distance travelled ?

- a) Tachometer
- b) Odometer.
- c) Speedometer
- d) Barometer

Learning Space



CAN SPEED BE USED IN BOTH CASES ??



CAR ON A HIGHWAY

Object covers equal distance
in equal intervals of time.

UNIFORM MOTION

SPEED



CARS ON A BUSY STREET

Object does not cover equal
distance in equal intervals of time.

NON-UNIFORM MOTION

AVERAGE SPEED



AVERAGE SPEED



30 meter in 20 seconds



30 meter in 10 seconds



$$\text{Average Speed} = \frac{\text{Total Distance Covered}}{\text{Total Time Taken}}$$

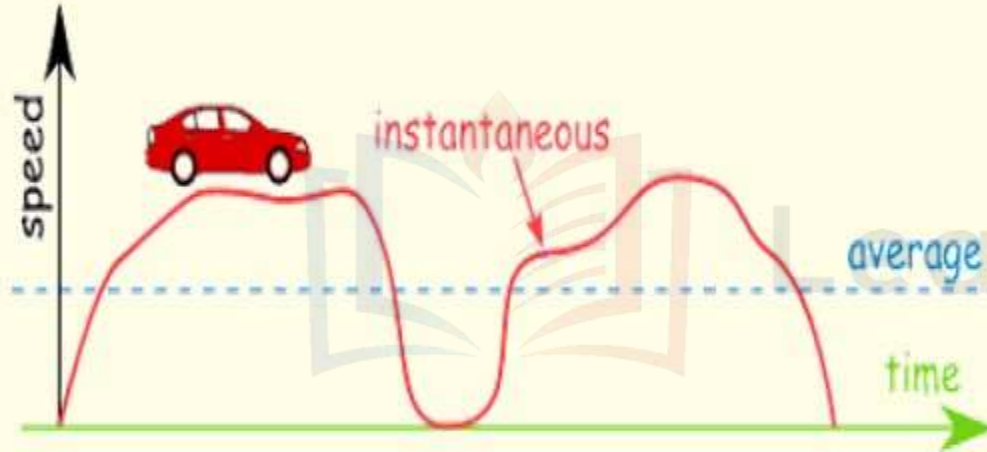
Total distance travelled by the boy is $(30 + 30) = 60 \text{ m}$

Total time taken by the boy $(20+10) = 30 \text{ s}$

Therefore, average speed is $(60/30) = 2 \text{ ms}^{-1}$



SPEED CALCULATION - DIFFERENT CASES



AVERAGE SPEED

Average Speed is defined as the **total distance covered** divided by **total time taken**

INSTANTANEOUS SPEED

Instantaneous Speed is the speed of a continuously changing object at any **particular instant**





SPEED - VECTOR QUANTITIES

SPEED

- Speed is a scalar quantity. It only has magnitude but no direction.
- Speed is defined as **distance** covered in unit time.
- Speed of any moving object is always positive.

VELOCITY

- Velocity is the vector quantity of speed. It has both magnitude and a definite direction.
- Velocity is defined as **displacement** covered in unit time.
- Velocity of any moving object can be positive, negative or zero !

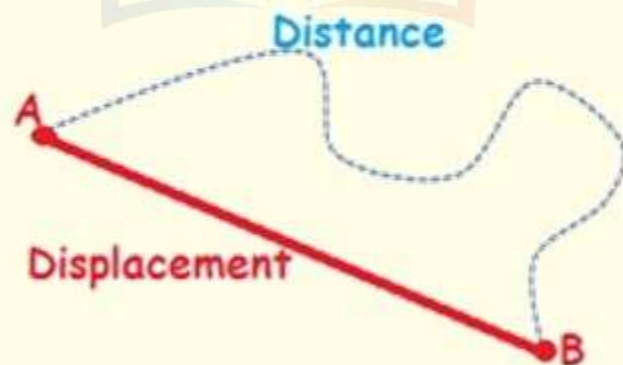
Any object moving in a circular path and coming back to the starting point has a speed but has 0 velocity ! Why ?



DISTANCE & DISPLACEMENT

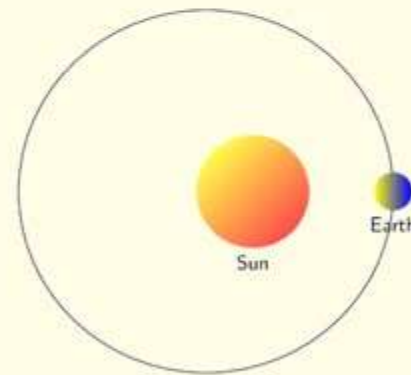
DISTANCE

- Length of a path travelled by a body.
- It is a Scalar quantity.
- Distance is always positive.



DISPLACEMENT

- Shortest distance travelled by a body.
- It is a vector quantity.
- It can be positive, negative or zero!



Any object moving in a circular path and coming back to the starting point covers a distance but has 0 displacement !



AVERAGE VELOCITY



30 meter in 20 seconds



30 meter in 10 seconds



$$\text{Average Velocity} = \frac{\text{Initial Velocity} + \text{Final Velocity}}{2}$$

**But, in this case average velocity is 0 !!!
Can you tell why ?**



AVERAGE VELOCITY



30 meter in 20 seconds



30 meter in 10 seconds



$$\text{Average Velocity} = \frac{\text{Initial Velocity} + \text{Final Velocity}}{2}$$

Displacement is 0 in this case as the boy is coming back to his original position. Therefore, Average Velocity also becomes zero !



**QUESTION**

When speed of object changes, the velocity -

- a) Remains same
- b) increases
- c) decreases
- d) Also changes.





QUESTION

Under what conditions the magnitude of average velocity of an object equal to the average speed?

- a) Distance travelled is greater than net displacement
- b) Distance travelled equals net displacement.
- c) Distance travelled is less than net displacement
- d) Average velocity and average speed can never be equal



RATE OF CHANGE OF VELOCITY



UNIFORM MOTION

CAR ON A HIGHWAY

Velocity **remains constant**
with time



NON-UNIFORM MOTION

CARS ON A BUSY STREET

Velocity **varies** with time



RATE OF CHANGE OF VELOCITY - ACCELERATION

ACCELERATION - DEFINITION

Acceleration is defined as the measure of change of velocity of an object per unit time.

$$\text{acceleration} = \frac{\text{Final velocity} - \text{initial velocity}}{\text{time}}$$

SI

Meter per second square inverse
 ms^{-2}

*Acceleration is taken as positive if it is the direction of velocity.
Acceleration is negative if it is opposite to the direction of velocity*

ACCELERATION SIMPLIFIED !

A boy starts from rest and paddles his bicycle to attain a velocity of 6 ms^{-1} in 30s. Then he applies brakes such that the velocity of the bicycle comes down to 4 ms^{-1} in 5s. Calculate the acceleration in both cases.



Initial velocity = 0

30 sec



Final velocity = 6 ms^{-1}
Initial velocity = 6 ms^{-1}

5 sec



Final velocity = 4 ms^{-1}





30 sec



5 sec



Initial velocity = 0

Final velocity = 6 ms^{-1}

Initial velocity = 6 ms^{-1}

Final velocity = 4 ms^{-1}

$$a = \frac{v-u}{t}$$

$$a = \frac{(6-0)}{30}$$

$$a = 0.2 \text{ ms}^{-2}$$

$$a = \frac{v-u}{t}$$

$$a = \frac{(4-6)}{5}$$

$$a = -0.4 \text{ ms}^{-2}$$



TYPES OF ACCELERATION



UNIFORM ACCELERATION

CAR ON A HIGHWAY

Velocity can increase or decrease at equal intervals of time

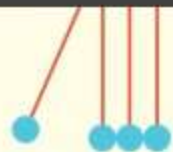


NON-UNIFORM ACCELERATION

CARS ON A BUSY STREET

Velocity can increase or decrease at unequal intervals of time





QUESTION

Which of the following are examples of uniform acceleration ?

- 1) A boy riding a bicycle in the same speed
- 2) An object falling from a cliff
- 3) A rocket lifts off from a space center
- 4) A race car at a constant speed in a circular track

Select from the codes given below;

- | | |
|-----------------|----------------|
| a) 1 & 2 only | b) 2,3 &4only. |
| c) 1,2,& 3 only | d) All 1,2,3,4 |

Velocity is a vector quantity of speed, it continuously changes when an object is moving in circles.



VECTOR QUANTITIES - RECAP

DISPLACEMENT

Shortest distance traversed by an object
Can be +ve, -ve and zero!
Eg. 5km - east, 10 km above



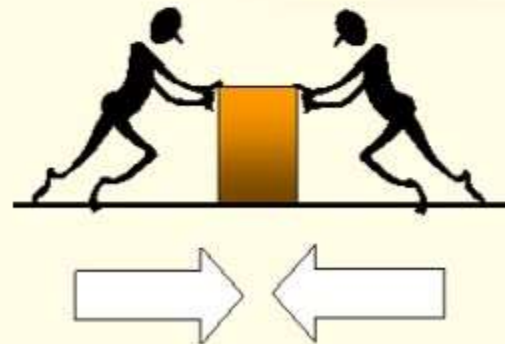
VELOCITY

Similar to Speed but has direction
Eg. 5 m/s up, 10 m/s in north



ACCELERATION

The rate of change of velocity which has a particular direction.
Negative acceleration is also called retardation.



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